

Project Design Phase-II

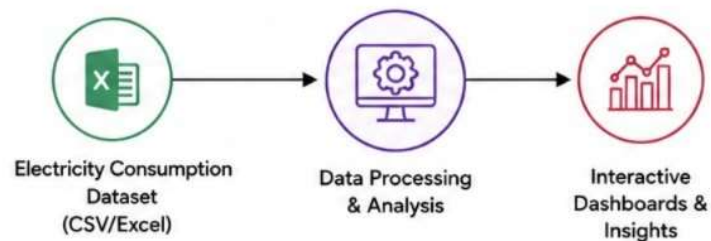
Data Flow Diagram & User Stories

Date	29 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Data Flow Diagrams:

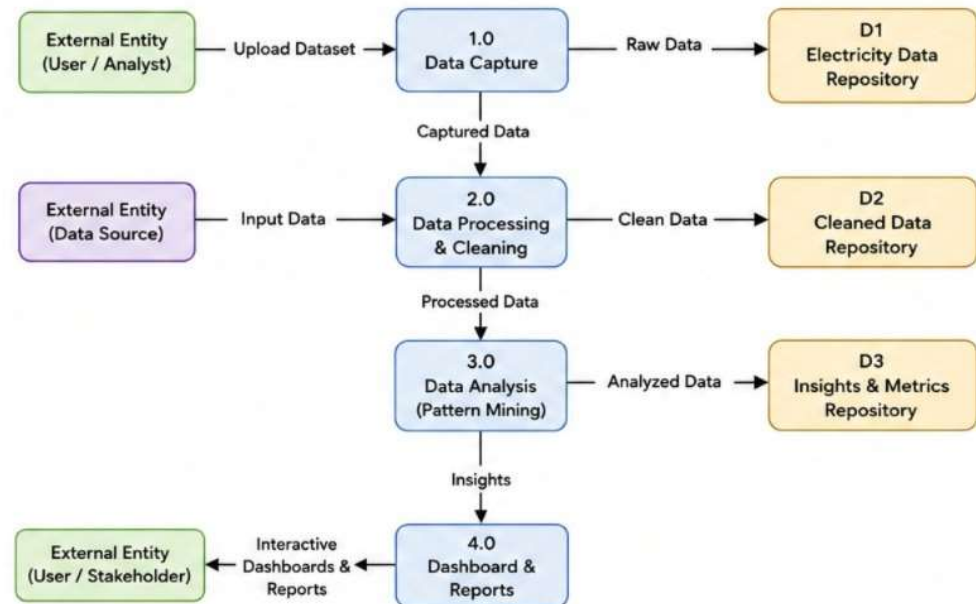
A Data Flow Diagram (DFD) is a graphical representation of how electricity consumption data moves through the system. It illustrates how data is collected from the dataset, processed for analysis, transformed into meaningful visualizations, and presented to users. The DFD clearly shows the flow of information, the processing steps, and how the analyzed data is stored and accessed to support better energy consumption analysis decision making.

Example: (Simplified)



1. User uploads the electricity consumption dataset (CSV/Excel).
2. Data is processed and cleaned.
3. Data is analyzed to find patterns and trends.
4. Insights (e.g., peak usage, cost analysis, comparisons) are generated.
5. Results are presented to users through interactive dashboards.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Data Analyst	Data Upload	USN-1	As a data analyst, I can upload the electricity consumption dataset into the system.	Dataset is uploaded successfully and ready for analysis.	High	Sprint-1
	Data Validation	USN-2	As a data analyst, I can validate the uploaded dataset for missing or incorrect values.	Invalid or missing data is identified and reported.	High	Sprint-1
	Data Processing	USN-3	As a data analyst, I can clean and prepare the dataset for visualization.	Cleaned data is stored and available for analysis.	High	Sprint-1
Dashboard User	Dashboard	USN-4	As a user, I can view an interactive dashboard showing electricity consumption trends.	Dashboard displays charts and KPIs correctly.	High	Sprint-1
	Trend Analysis	USN-5	As a user, I can analyze monthly and yearly electricity consumption trends.	Trend charts update based on selected time period.	High	Sprint-1
	State-wise Analysis	USN-6	As a user, I can compare electricity consumption across different states.	State-wise comparison charts are displayed accurately.	High	Sprint-2
	COVID-19 Impact Analysis	USN-7	As a user, I can analyze electricity consumption before, during, and after the COVID-19 lockdown.	Dashboard displays comparisons for different periods.	Medium	Sprint-2
	Filter Data	USN-8	As a user, I can filter the dashboard by state, month, and year.	Visualizations update dynamically based on selected filters.	High	Sprint-2
	Reports	USN-9	As a user, I can download reports and dashboard summaries.	Reports are generated successfully in PDF or image format.	Medium	Sprint-3

Administrator	Data Management	USN-10		As an administrator, I can update or replace the electricity consumption dataset.	New dataset is uploaded without affecting the dashboard.	High	Sprint-3
	User Management	USN-11		As an administrator, I can manage user access to the dashboard.	Authorized users can access the dashboard securely.	Medium	Sprint-3